

SKILLVANE IT ACADEMY · 2027 CURRICULUM

PROFESSIONAL CAREER PROGRAM

Multi-Cloud Data Engineer Program (GCP + Azure)

A production-style 4-month program with case studies and real-world projects. Build batch, streaming, and lakehouse pipelines the way industry teams do.

DURATION

4 months live sessions

LANGUAGE

English

MODE

Online live ·
8:00 PM to 9:00 PM

TRAINER

Shaik Saidhul

FEE

₹22,000
~~₹25,000~~

FREE DEMO

26 & 27 Sep
2027

Your 4-month learning journey

MONTH 1

Foundations

Cloud, GCP IAM, Linux on Cloud Shell, and Python for data engineers.

MONTH 2

GCP data platform

GCS, Cloud SQL + DMS, BigQuery analytics, and PySpark on Dataproc.

MONTH 3

Streaming & Azure

Dataflow, Pub/Sub, Composer DAGs, Azure foundations, and ADLS Gen2.

MONTH 4

Lakehouse & projects

ADF, Synapse, Databricks Delta/Unity Catalog, and interview-ready capstones.

Tools you will master

SQL

Python

Linux

PySpark

Apache Beam

Airflow

BigQuery

Dataproc

Dataflow

Pub/Sub

Composer

ADLS Gen2

Azure Data Factory

Synapse

Databricks

Delta Lake

Structured curriculum

01 · Cloud Computing Foundations

- What cloud computing is and how it supports modern businesses
- Why organizations moved from on-premises infrastructure to cloud
- Cloud service models and shared-responsibility fundamentals
- Cloud providers and ecosystems: Google Cloud, Microsoft Azure, and AWS
- Data Lake, Data Warehouse, and Lakehouse architecture concepts

02 · GCP Platform, IAM & Linux

- GCP core services and interfaces: Console, Cloud Shell, and Cloud SDK
- Regions, zones, global infrastructure, and location selection
- Organizations, folders, projects, billing accounts, and resource hierarchy
- IAM principals, roles, policies, service accounts, and least privilege
- Step-by-step GCP free-account setup and Cloud Shell workflow
- Linux installation, terminal navigation, and Cloud Shell productivity tips
- File operations: ls, cd, pwd, mkdir, cp, mv, touch, rm, and nano
- Content and text processing: cat, less, head, tail, grep, awk, sed, cut, sort, uniq
- Users and permissions: whoami, id, su, sudo, chmod, and chown

03 · Python for Data Engineers

- Python setup, basic functions, strings, numbers, booleans, and operators
- Lists, tuples, dictionaries, sets, and common data manipulation patterns
- Conditional logic with if, elif, and else; for and while loops
- File I/O: read, write, append, rename, remove, copy, and move
- Exception handling with try, except, else, finally, and raise
- Production logging, log levels, file handlers, and best practices
- Date and time operations: formatting, parsing, and timedeltas
- Modular programming, functions, lambda functions, and classes
- REST API fundamentals, HTTP methods, status codes, headers, and JSON
- Calling and reading API responses with the requests library
- Modules, packages, imports, built-in modules, and package installation

04 · Google Cloud Storage

- Scalable object storage concepts, buckets, objects, and common use cases
- Console and CLI labs for creating and managing storage buckets
- Uploading and downloading objects through multiple interfaces
- Bucket and object access controls, permissions, and security
- Data transfer methods and lifecycle-management policies
- Object versioning, retention, recovery, and cost optimization
- Integration with BigQuery, Dataproc, Dataflow, and other GCP services
- Monitoring and logging storage operations

05 · Cloud SQL & Database Migration

- Cloud SQL architecture, supported engines, and common use cases
- Creating instances and configuring databases, users, and access controls
- Connecting with Cloud SQL Studio, Cloud Shell, and database workbenches
- Importing, exporting, backup configuration, and high availability
- Integration with GCP data services and database role management
- Database Migration Service fundamentals and migration planning
- End-to-end project: manual export and import migration
- End-to-end project: automated migration using Cloud SQL DMS

06 · BigQuery SQL & Analytics

- BigQuery architecture, serverless analytics, and business-intelligence use cases
- Table-creation methods, supported sources, and file formats
- Native and external tables and working with complex data types
- JSON, nested, repeated, STRUCT, and ARRAY data
- Loading from Cloud Storage, Cloud SQL, and other sources
- Exporting data and implementing real-time streaming ingestion
- IAM and access controls for datasets, tables, and jobs
- Standard, materialized, and authorized views
- Optimization with partitioning, clustering, query plans, and cost controls
- BigQuery slots: on-demand, flat-rate, and flex-slot concepts
- Case study: real-world Spotify analytics data platform
- Case study: enterprise social-media analytics platform

07 · Dataproc & PySpark

- Hadoop, Spark, MapReduce, and the PySpark execution model
- Dataproc as a managed Spark and Hadoop platform
- Cluster creation, properties, performance, scalability, and preemptible VMs
- Reading CSV, text, JSON, Parquet, database, and BigQuery sources
- DataFrame transformations, cleansing, enrichment, and aggregation
- Writing to files, databases, and BigQuery targets
- Submitting and monitoring Spark jobs on Dataproc
- Initialization actions, custom scripts, debugging, and troubleshooting
- Case study: employee travel-record data cleaning
- Case study: processing real-time patient health data
- Case study: PySpark feature-processing job for ML model creation

08 · Dataflow, Apache Beam & Pub/Sub

- Dataflow architecture and batch versus streaming use cases
- Apache Spark versus Apache Beam; Dataproc versus Dataflow
- Beam pipelines, PCollections, transforms, runners, and I/O connectors
- Reading and writing files, databases, and BigQuery datasets
- Template-based pipeline development and deployment
- Pub/Sub event-driven architecture, topics, subscriptions, and messages
- Retention policies, acknowledgements, retries, and delivery behavior
- Publishing and consuming messages with Python applications
- Case study: e-commerce transaction processing
- Case study: end-to-end Python + Pub/Sub + Dataflow + BigQuery + GCS pipeline

09 · Cloud Composer (Airflow DAGs)

- Apache Airflow architecture and Cloud Composer use cases
- Creating and configuring managed Composer environments
- Defining, scheduling, monitoring, and managing DAG executions
- Orchestrating BigQuery, Dataproc, Dataflow, external systems, and APIs
- ETL dependencies, retries, error handling, logging, and troubleshooting
- Level 1 DAG: orchestrating BigQuery pipelines
- Level 2 DAG: orchestrating Dataproc pipelines
- Level 3 DAG: orchestrating Dataflow pipelines
- CI/CD deployment using Cloud Build and GitHub

10 · Azure Foundations & ADLS Gen2

- Azure core services, Portal, Cloud Shell, CLI, SDK, regions, and zones
- Subscriptions, resources, resource groups, and management groups
- Azure identities, roles, service principals, hierarchy, and free-account setup
- ADLS Gen2 storage accounts, containers, blobs, and hierarchical namespace
- Scalability, security, cost effectiveness, and analytics integrations
- Data management using Portal, SDK, CLI, and REST APIs
- Role-Based Access Control and Access Control Lists
- Hot, Cool, and Archive tiers and storage cost factors
- Partitioning, tiering, monitoring, and usage-tracking best practices

11 · Azure SQL, Data Factory & Synapse

- Azure SQL Database concepts, features, provisioning, and practice setup
- ADF use cases and core components: integration runtimes and linked services
- Datasets, pipelines, activities, parameters, and triggers
- Creating, scheduling, monitoring, and managing pipeline runs
- Orchestration patterns using multiple ADF activities
- Efficient pipeline design, partitioning, parallelism, and optimization
- Case study: end-to-end incremental ADF + ADLS + Azure SQL pipeline
- Synapse workspaces, architecture, dedicated SQL, and serverless SQL pools
- Data ingestion with Synapse Pipelines
- External tables, views, and Data Lake integration
- Query optimization, partitioning, and workload management

12 · Databricks Lakehouse (Azure / GCP)

- Lakehouse architecture and Databricks workspace, cluster, and job components
- Databricks Runtime, Apache Spark, notebooks, collaboration, and autoscaling
- Workspace administration, user management, permissions, and billing
- Unity Catalog for metadata, governance, access control, and secure sharing
- Delta Lake ACID transactions, versioning, time travel, and table management
- Creating managed and external Delta tables across multiple formats
- Building ELT pipelines with Spark SQL and Python
- Incremental data processing and production workflow design
- Case study: real-time processing with Spark Structured Streaming
- Case study: file ingestion using Databricks Auto Loader
- Performance optimization for Spark and Delta workloads

Flagship case studies & real-world projects

BIGQUERY Spotify analytics platform

Design a real-world analytics warehouse with nested data, views, and governed access.

BIGQUERY Enterprise social media analytics

Ingest, model, and serve campaign-scale social data for business intelligence.

PYSPARK Travel, healthcare & ML feature jobs

Clean employee travel records, process patient health data, and prepare ML datasets.

STREAMING E-commerce + Pub/Sub + Dataflow

Ship an end-to-end streaming pipeline across Python, Pub/Sub, Dataflow, BigQuery, and GCS.

AZURE Incremental ADF pipeline

Build production-style incremental load with ADF, ADLS, and Azure SQL Database.

DATABRICKS Streaming + Autoloader

Process real-time data with Structured Streaming and ingest files with Autoloader.

What students can expect

- Build end-to-end data engineering pipelines on GCP and Azure
- Develop strong SQL, Python, PySpark, Linux, and Apache Beam skills
- Implement batch and real-time processing pipelines
- Design Bronze → Silver → Gold / Medallion architecture
- Work with industry-standard DE services and lakehouse patterns
- Ship production-ready projects and troubleshoot real scenarios
- Understand cloud, data lake, warehouse, and lakehouse architectures
- Prepare for data engineer interviews with scenario-based practice

Course highlights

Live instructor-led sessions for 4 months

Recordings shared daily

Comprehensive study material and code resources

Hands-on labs and real-world case studies

Live doubt-clearing and Q&A sessions

Interview preparation and career guidance

Shaik Saidhul · Cloud Expert · SkillVane IT Academy

Teaching language: English · Mode: Online · Phone/WhatsApp: +91 7305101711

Thank you. Learn · Build · Advance.